UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 2:
IDENTIFYING AND HANDLING MISSING DATA WITH SPREADSHEETS

DATA WRANGLING AND VISUALIZATION (SAIA2143)

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TOPIC LEARNING OUTCOMES

At the end of this class, students should be able to:

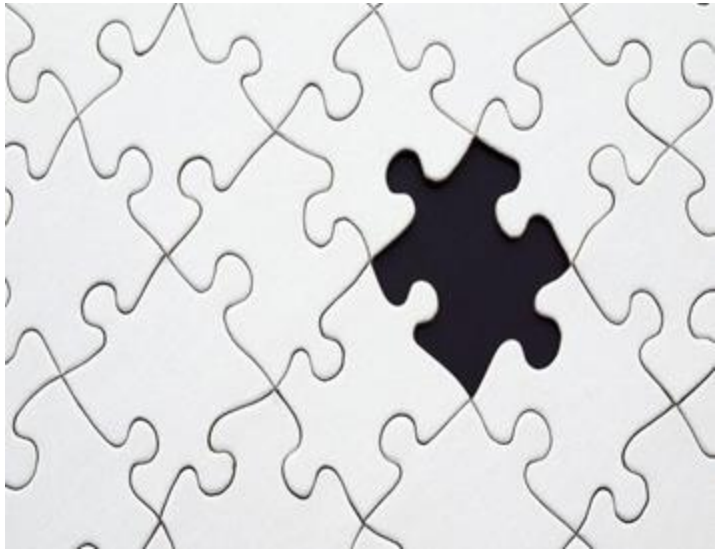
1. Identify the importance of handling missing data
2. Explain sorting, filtering, and basic data cleaning techniques
3. Identify missing values in datasets
4. Apply spreadsheet tools to manage missing data

Introduction to Missing Data

Definition

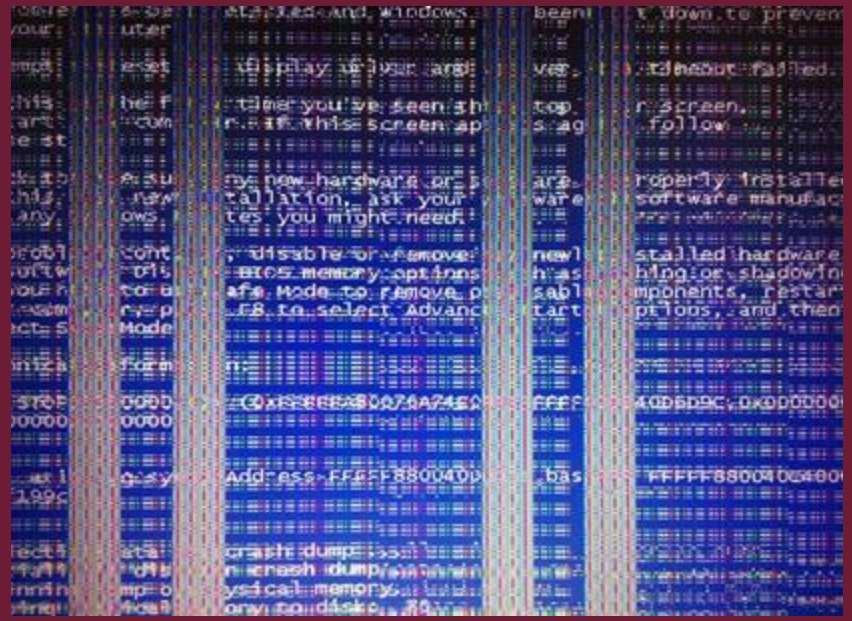
- Missing data refers to the absence of values in a dataset where information is expected.

- It can occur due to various reasons, including errors during data collection, transmission, or storage.



Causes of Missing Data

- **Human Error:** Mistakes in data entry, survey non-responses, or skipped fields.
- **System Glitches:** Software bugs, data corruption, or power failures.
- **Data Processing Issues:** Loss during extraction, transformation, or merging of datasets.
- **Privacy & Compliance:** Intentional data omission due to confidentiality concerns.



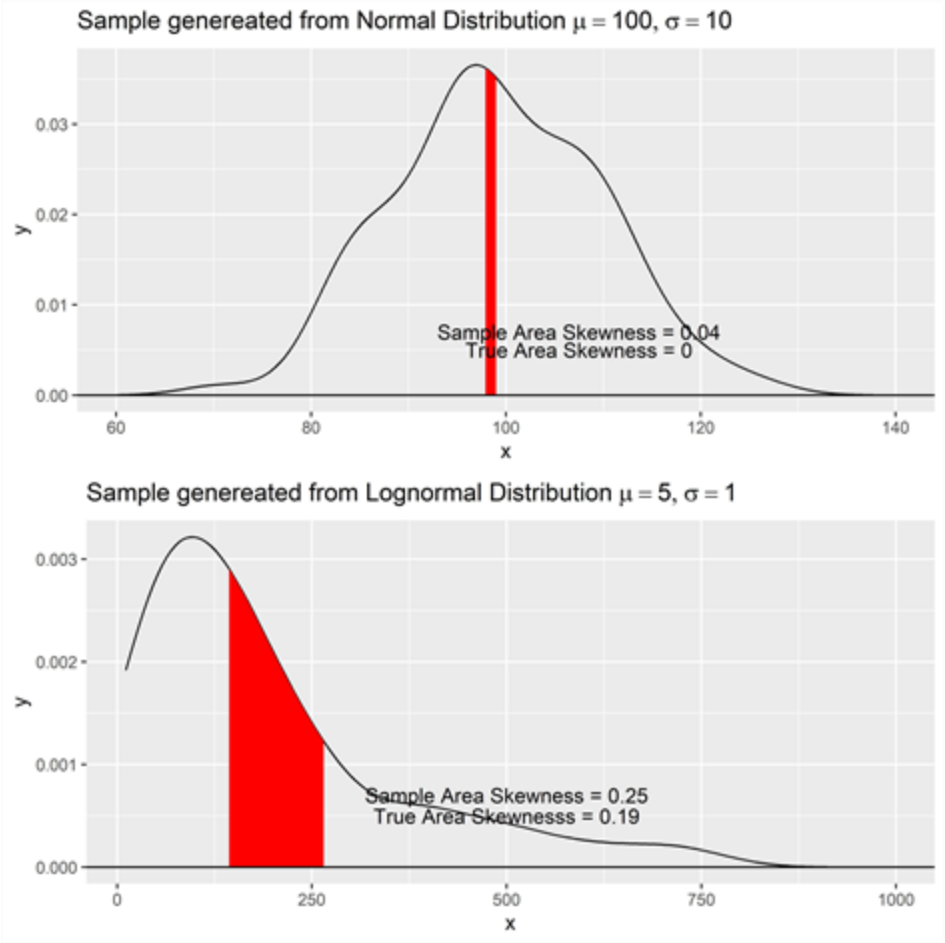
Impact of Missing Data

Example of Missing data	Impact
Bias in Training Data	Missing values can distort statistical findings.
Loss of Data Integrity	Reduced completeness lowers data reliability.
Misleading Insights	Inaccurate conclusions may lead to poor decision-making.
Challenges in Model Performance	Machine learning models may struggle with incomplete data.

a. Bias in Training Data

- If missing data is not randomly distributed (i.e., missing systematically), the model may learn biased patterns.

Example: If customer income data is missing mostly for lower-income individuals, the model may wrongly assume all customers have higher incomes, leading to skewed predictions.



b. Loss of Data Integrity

- Data integrity refers to the accuracy, consistency, and reliability of data throughout its lifecycle. When data is incomplete or missing, its integrity is compromised, making it unreliable for analysis and decision-making.

Example Decision-Making : A hospital's patient database missing medical history records can lead to incorrect treatment plans, putting patients at risk.

c. Misleading Insights

- When data is incomplete or missing, it can distort analysis and lead to incorrect conclusions. This can significantly impact decision-making in various fields.

Example Marketing Campaign Failures: A company launches a product targeting a specific audience, but due to missing customer preference data, the campaign fails, wasting resources.

d. Challenges in Model Performance

- Some algorithms, especially those relying on gradient-based optimization (e.g., neural networks), struggle with missing data because: Large gaps in data create unstable weight updates.
- Training may fail to converge to an optimal solution.

Example: In deep learning, missing pixel values in image classification may disrupt convolutional layers, leading to incorrect feature extraction.



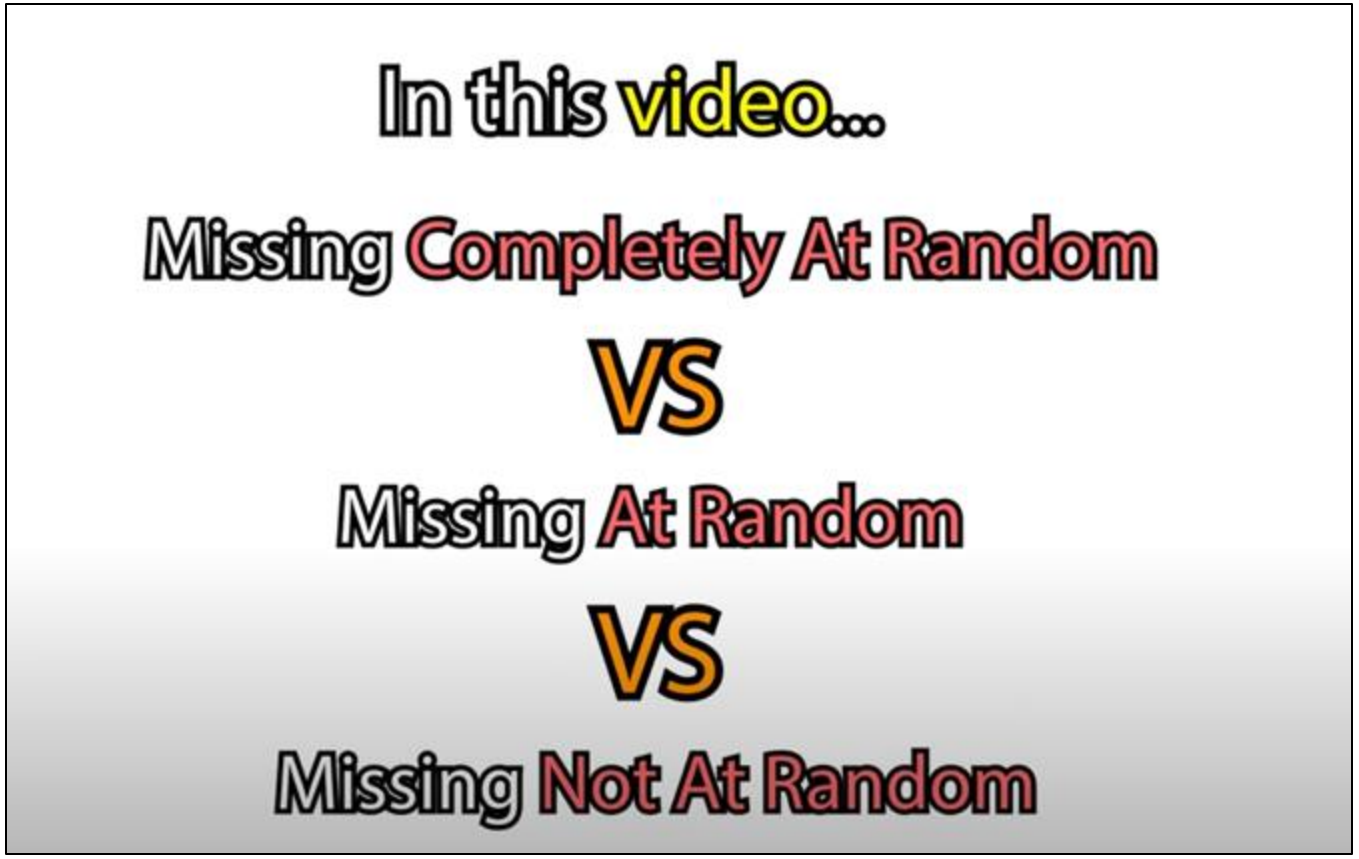
How is a Missing Value Represented in a Dataset?

1. **NaN (Not a Number)**: In many programming languages and data analysis tools, missing values are represented as NaN. This is the default for libraries like **Pandas** in Python.
2. **NULL or None**: In databases and some programming languages, missing values are often represented as NULL or None. For instance, in **SQL databases**, a missing value is typically recorded as NULL.
3. **Empty Strings**: Sometimes, missing values are denoted by empty strings (""). This is common in text-based data or CSV files where a field might be left blank.
4. **Special Indicators**: Datasets might use specific indicators like -999, 9999, or other unlikely values to signify missing data. This is often seen in older datasets or specific industries where such conventions were established.
5. **Blanks or Spaces**: In some cases, particularly in fixed-width text files, missing values might be represented by spaces or blank fields.

Handling Strategies of Missing Data

- **Data Imputation:** Filling missing values with estimated or default values.
- **Data Cleaning:** Removing incomplete records when necessary.
- **Advanced Techniques:** Using predictive modelling to estimate missing values.

3 Types of Missing Data



YouTube

Instruction: Please listen the YouTube Video and answer the question. (Duration: 3:06 minutes)

https://www.youtube.com/watch?v=UzsWr9X98J8

- Understanding the type of missing data is crucial because different types require different handling strategies.
- Ignoring MNAR data can create large biases

Missing Completely At Random (MCAR)

MCAR occurs when the probability of data being missing is uniform across all observations. There is no relationship between the missingness of data and any other observed or unobserved data within the dataset. This type of missing data is purely random and lacks any discernible pattern.

Example: In a survey about library books, some overdue book values in the dataset are missing due to human error in recording.

Missing At Random (MAR)

MAR data occurs when the probability of data being missing depends only on the observed data and not on the missing data itself. In other words, the missingness can be explained by variables for which you have complete information. There is a pattern in the missing values, but this pattern can be explained by other observed variables.

Example: In a survey, 'Age' values might be missing for those who did not disclose their 'Gender'. Here, the missingness of 'Age' depends on 'Gender', but the missing 'Age' values are random among those who did not disclose their 'Gender'.

Missing Not At Random (MNAR)

MNAR occurs when the missingness of data is related to the unobserved data itself, which is not included in the dataset. This type of missing data has a specific pattern that cannot be explained by observed variables.

Example: In a survey about library books, people with more overdue books might be less likely to respond to the survey. Thus, the number of overdue books is missing and depends on the number of books overdue.



CLASS ACTIVITY

Instructions: Match each scenario to the correct type of missing data;

MCAR (Missing Completely at Random), MAR (Missing at Random), or MNAR (Missing Not at Random).

Scenarios:

1. Customers who are dissatisfied with a product do not leave reviews, resulting in an overly positive rating. (____)
2. A researcher's clipboard gets wet, causing some survey responses to be unreadable. (____)
3. Patients who are older tend to skip questions about their physical activity levels, but younger respondents provide answers. (____)
4. A temperature sensor in a factory occasionally fails to record data due to random malfunctions. (____)
5. Employees with higher education levels are more likely to complete an annual job satisfaction survey. (____)
6. Students who perform poorly on an exam are less likely to report their scores in a voluntary survey. (____)

The Role of Spreadsheets in Handling Missing Data

- **Accessibility**: Widely available and easy to use.
- **Flexibility**: Can handle both small and large datasets.
- **Visualization**: Allows for quick identification of missing values.
- **Built-in Functions**: Includes various tools to clean and analyze data.
- **Integration**: Compatible with other data processing tools.

Overview of Tools Available in Excel/Google Sheets

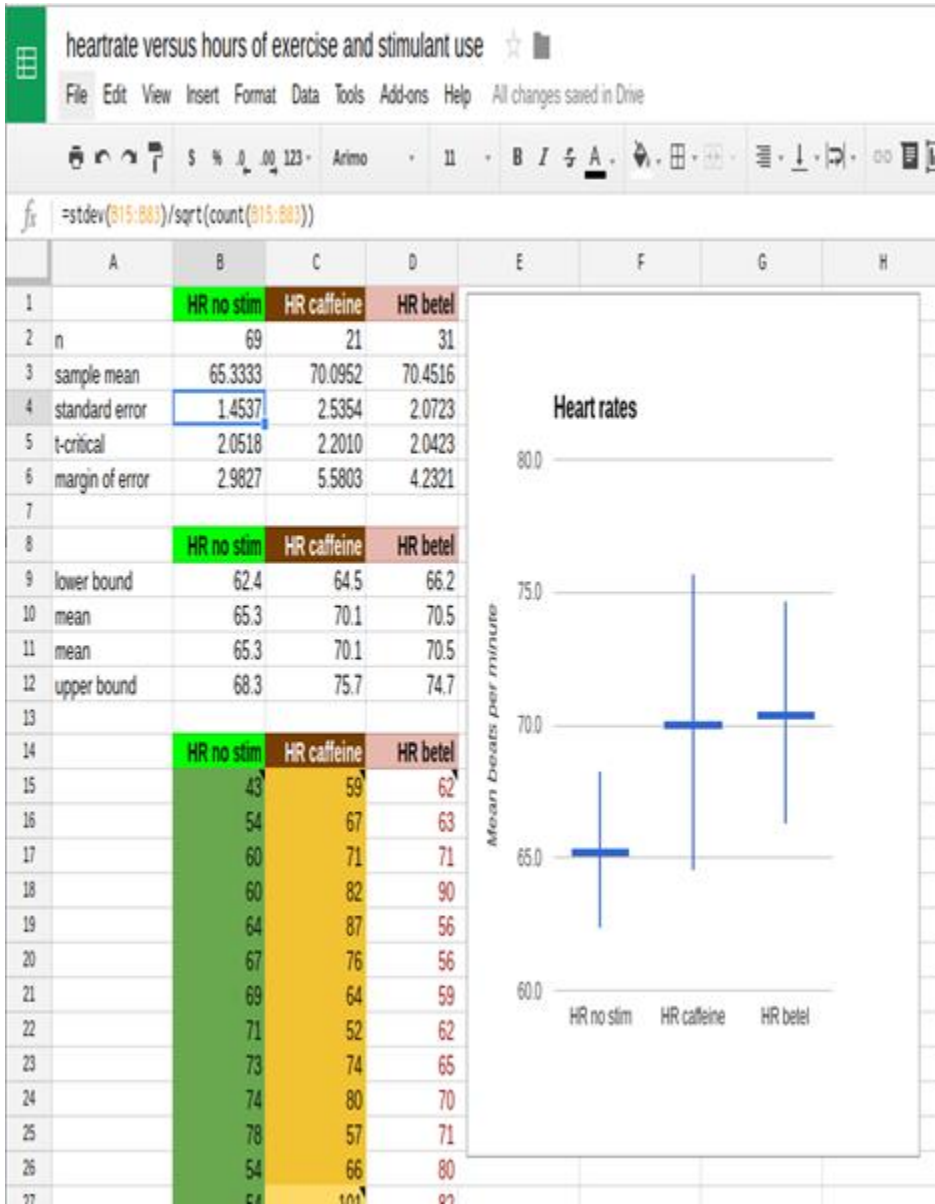
a. **Sorting & Filtering**: Identify and remove empty or duplicate values.

b. **Conditional Formatting**: Highlight missing or inconsistent data.

c. **Find & Replace**: Locate and correct missing values efficiently.

d. **Data Validation**: Prevent missing or incorrect entries.

e. **Pivot Tables**: Summarize and analyze missing data trends.



	A	B	C	D
1		HR no stim	HR caffeine	HR betel
2	n	69	21	31
3	sample mean	65.3333	70.0952	70.4516
4	standard error	1.4537	2.5354	2.0723
5	t-critical	2.0518	2.2010	2.0423
6	margin of error	2.9827	5.5803	4.2321
7				
8		HR no stim	HR caffeine	HR betel
9	lower bound	62.4	64.5	66.2
10	mean	65.3	70.1	70.5
11	mean	65.3	70.1	70.5
12	upper bound	68.3	75.7	74.7
13				
14		HR no stim	HR caffeine	HR betel
15		43	59	62
16		54	67	63
17		60	71	71
18		60	82	90
19		64	87	56
20		67	76	56
21		69	64	59
22		71	52	62
23		73	74	65
24		74	80	70
25		78	57	71
26		54	66	80

a. Sorting and Filtering

File Name: ExerciseTopic2

Dataset: Customer Order

Order ID	Customer Name	Product	Quantity	Price ($)	Total ($)
1001	Alice Smith	Laptop	1	900	900
1002	Bob Johnson	Mouse	3	20	60
1003	Charlie Lee	Keyboard	2	50	100
1004	Alice Smith	Monitor	1	300	300
1005	David Kim	Laptop	1	950	950
1006	Emma Watson	Printer	2	120	240
1007	Frank Martin	Monitor	1	280	280
1008	Grace Hall	Mouse	5	18	90
1009	Henry Clark	Keyboard	3	55	165
1010	Alice Smith	Laptop	1	890	890
1011	Bob Johnson	Printer	1	125	125
1012	Charlie Lee	Monitor	2	310	620
1013	David Kim	Mouse	4	22	88
1014	Emma Watson	Laptop	1	930	930
1015	Frank Martin	Keyboard	2	50	100

Instructions: Use sorting and filtering functions in Excel or Google Sheets to manipulate this data and answer questions based on the sorted and filtered results.

1. Sorting:

a) **Sort by Total ($) in descending order** to identify the highest-value orders first.

b) **Sort by Customer Name alphabetically** to group purchases by each customer.

2. Filtering:

a) **Filter by Product = "Laptop"** to see all laptop purchases.

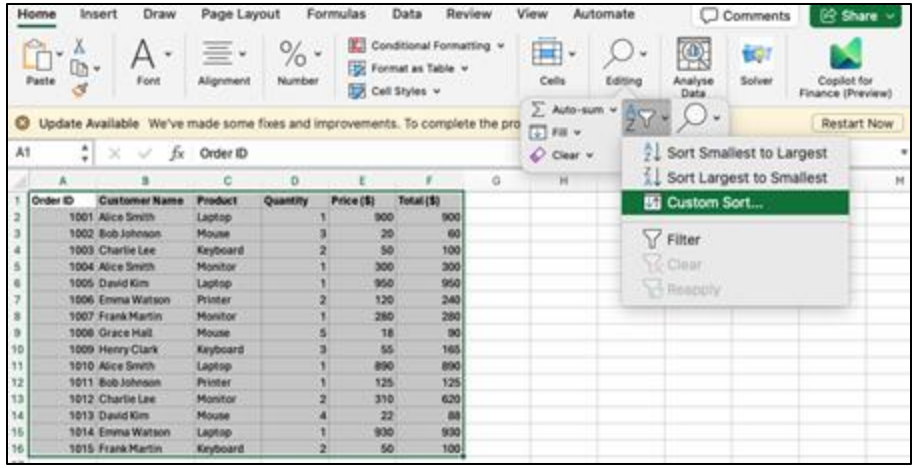
b) **Filter by Price > 100** to identify high-value product sales.

c) **Filter by Customer Name = "Alice Smith"** to view her purchases.

www.fai.utm.my

Sorting:

a) Sort by Total ($) in descending order to identify the highest-value orders first.



a. Highlight the area
b. Editing - Sort & Filter
c. Custom Sort

Answer:

Order ID	Customer Name	Product	Quantity	Price ($)	Total ($)
1005	David Kim	Laptop	1	950	950
1014	Emma Watson	Laptop	1	930	930
1001	Alice Smith	Laptop	1	900	900
1010	Alice Smith	Laptop	1	890	890
1012	Charlie Lee	Monitor	2	310	620
1004	Alice Smith	Monitor	1	300	300
1007	Frank Martin	Monitor	1	280	280
1006	Emma Watson	Printer	2	120	240
1009	Henry Clark	Keyboard	3	55	165
1011	Bob Johnson	Printer	1	125	125
1003	Charlie Lee	Keyboard	2	50	100
1015	Frank Martin	Keyboard	2	50	100
1008	Grace Hall	Mouse	5	18	90
1013	David Kim	Mouse	4	22	88
1002	Bob Johnson	Mouse	3	20	60

Sorting:

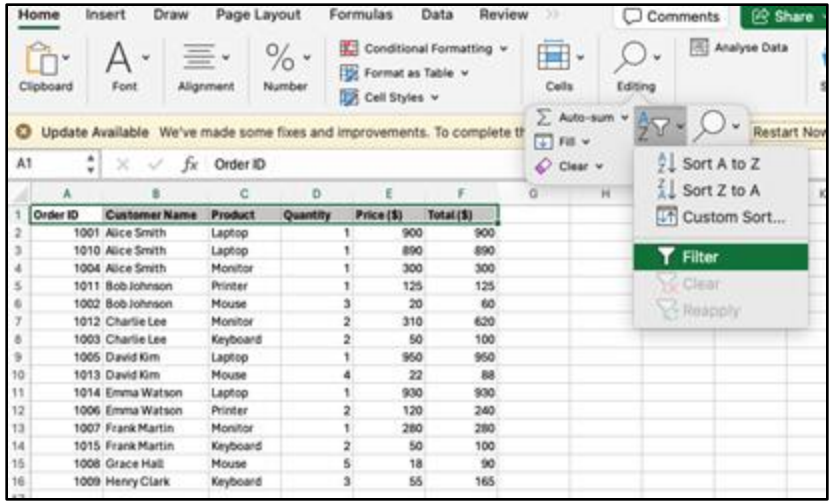
b) Sort by Customer Name alphabetically to group purchases by each customer.

A	B	C	D	E	F
Order ID	**Customer Name**	**Product**	**Quantity**	**Price ($)**	**Total ($)**
1001	Alice Smith	Laptop	1	900	900
1010	Alice Smith	Laptop	1	890	890
1004	Alice Smith	Monitor	1	300	300
1011	Bob Johnson	Printer	1	125	125
1002	Bob Johnson	Mouse	3	20	60
1012	Charlie Lee	Monitor	2	310	620
1003	Charlie Lee	Keyboard	2	50	100
1005	David Kim	Laptop	1	950	950
1013	David Kim	Mouse	4	22	88
1014	Emma Watson	Laptop	1	930	930
1006	Emma Watson	Printer	2	120	240
1007	Frank Martin	Monitor	1	280	280
1015	Frank Martin	Keyboard	2	50	100
1008	Grace Hall	Mouse	5	18	90
1009	Henry Clark	Keyboard	3	55	165

Answer:

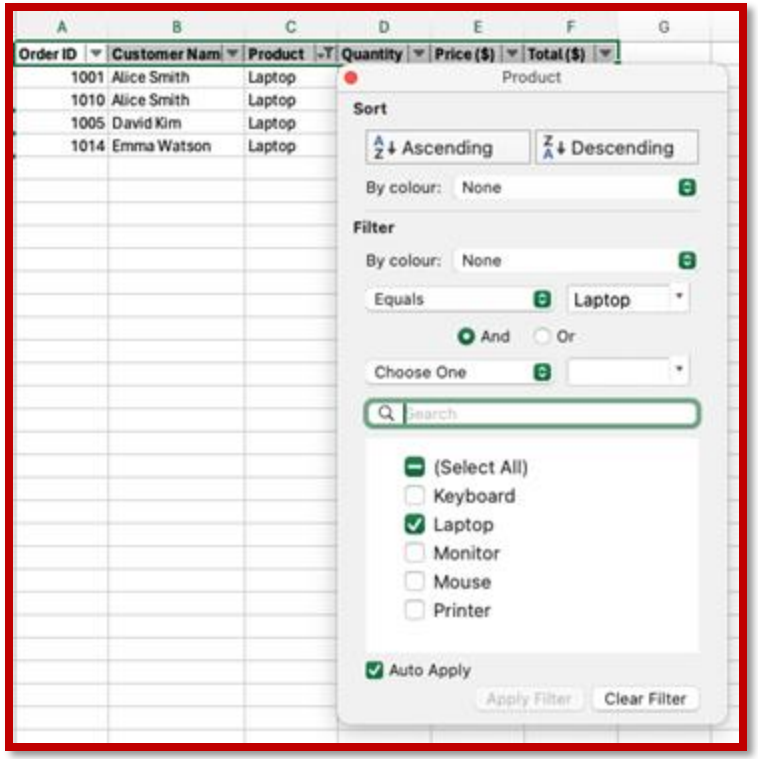
Filtering:

a) Filter by Product = "Laptop" to see all laptop purchases.



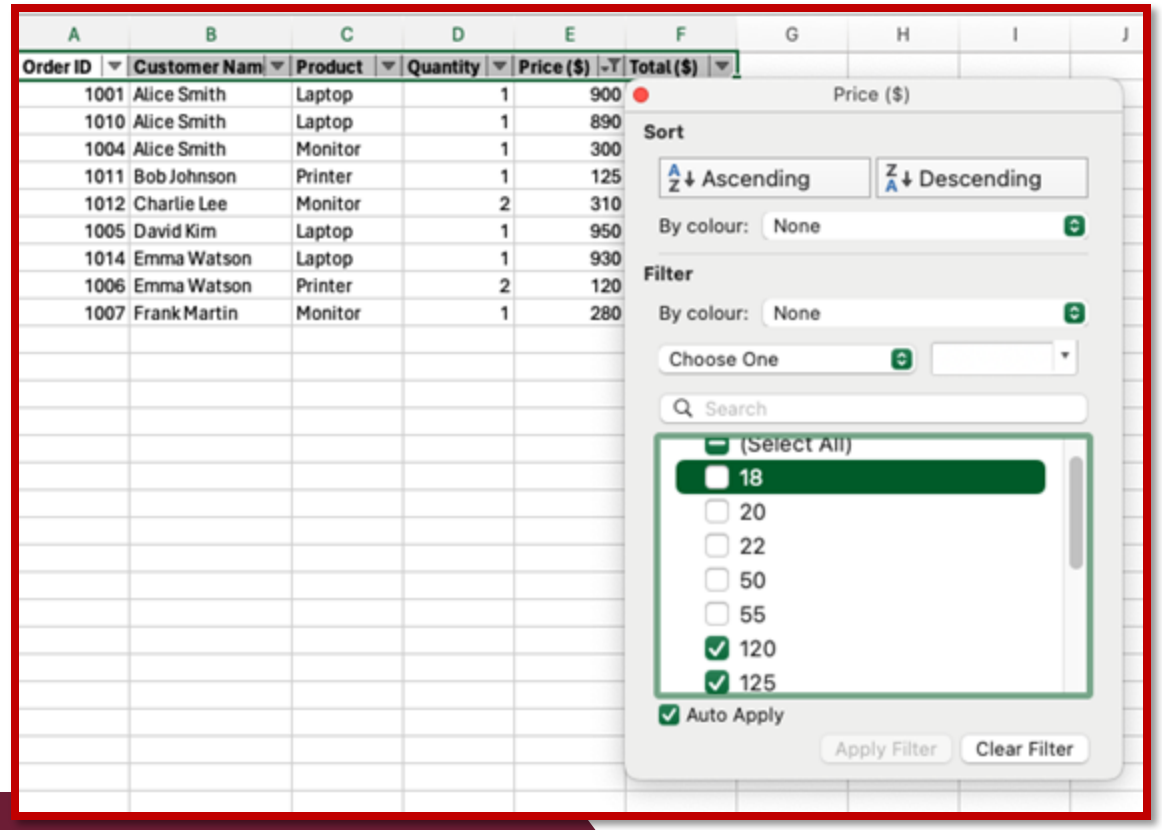
a. Highlight the first row
b. Editing - Sort & Filter
c. Filter

Answer:



Filtering:

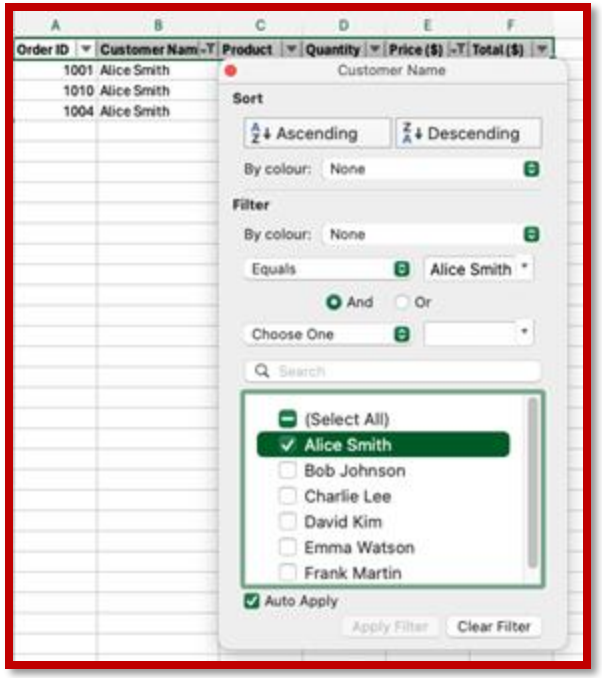
b) Filter by Price > 100 to identify high-value product sales.



Answer:

Filtering:

c) Filter by Customer Name = "Alice Smith" to view her purchases.

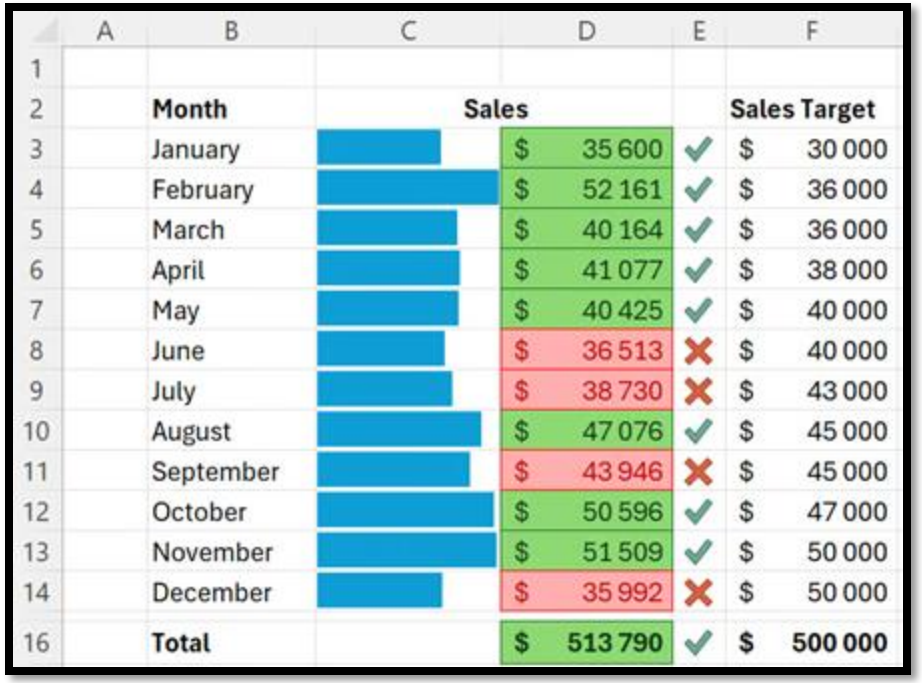


Answer:

Order ID	Customer Nam	Product	Quantity	Price ($)	Total ($)
1001	Alice Smith	Laptop	1	900	900
1010	Alice Smith	Laptop	1	890	890
1004	Alice Smith	Monitor	1	300	300

b. Conditional Formatting

- Allows users to automatically apply formatting (e.g., colors, bold text, icons) to cells based on specific conditions or criteria.
- It helps visually highlight key data trends, anomalies, or outliers.
- Types of conditional formatting:
  1. Highlight cells rules
  2. Top/bottom rules
  3. Color scales
  4. Data bars
  5. Icon sets



	A	B	C	D	E	F
1						
2		**Month**	**Sales**			**Sales Target**
3		January		$ 35 600	✓	$ 30 000
4		February		$ 52 161	✓	$ 36 000
5		March		$ 40 164	✓	$ 36 000
6		April		$ 41 077	✓	$ 38 000
7		May		$ 40 425	✓	$ 40 000
8		June		$ 36 513	✗	$ 40 000
9		July		$ 38 730	✗	$ 43 000
10		August		$ 47 076	✓	$ 45 000
11		September		$ 43 946	✗	$ 45 000
12		October		$ 50 596	✓	$ 47 000
13		November		$ 51 509	✓	$ 50 000
14		December		$ 35 992	✗	$ 50 000
16		**Total**		**$ 513 790**	✓	**$ 500 000**

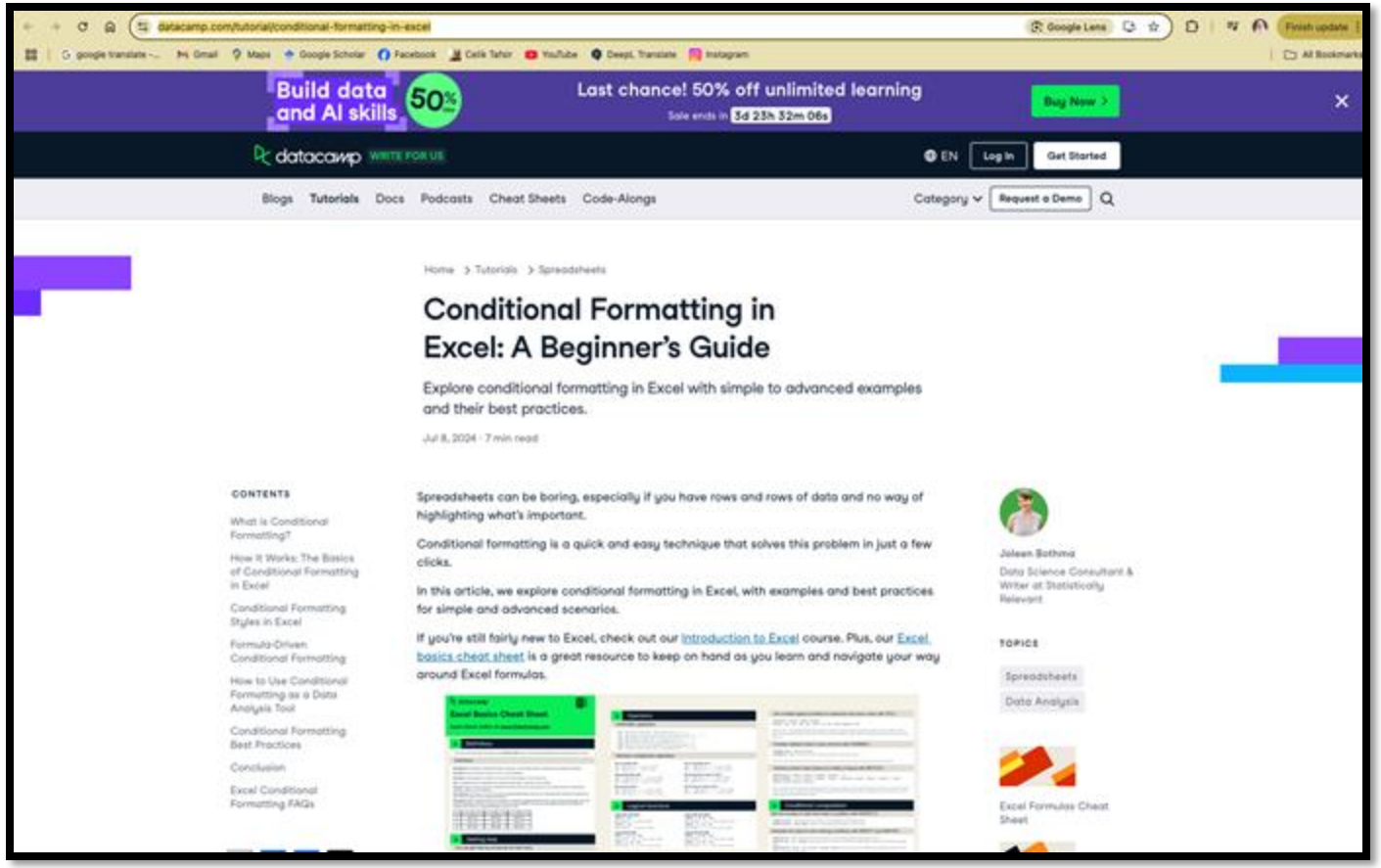


YouTube

9 minutes video

Link: https://www.youtube.com/watch?v=XHT4paRaY4g

CLASS ACTIVITY



Instruction:

According to the websites, what are Data Bars in Conditional Formatting, and how are they useful?

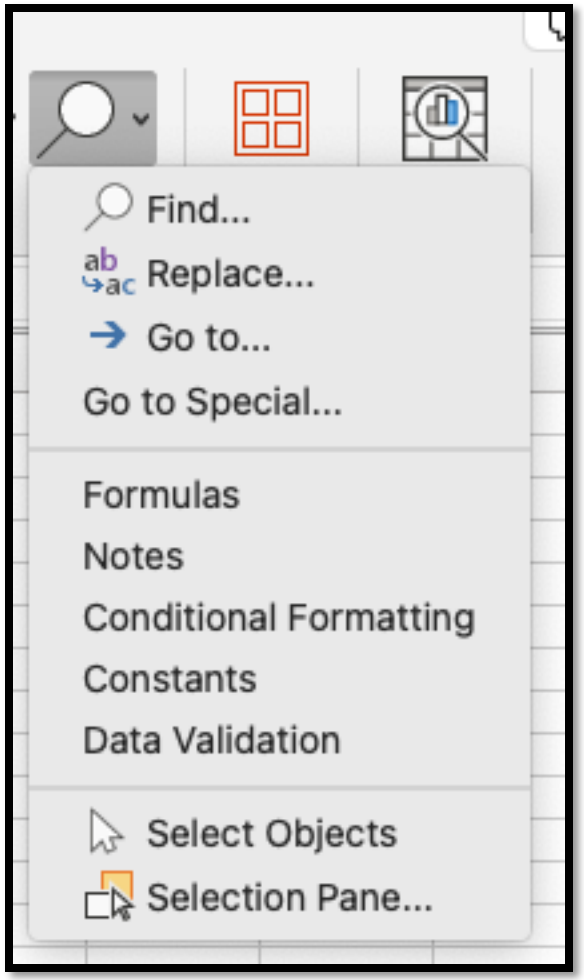
https://www.datacamp.com/tutorial/conditional-formatting-in-excel

www.fai.utm.my

c. Find & Replace

- Helps users quickly locate specific text, numbers, or formatting in a worksheet and replace them with new content.

- It is especially useful when working with large datasets that require bulk modifications or corrections.



Use the Find and Replace tool to standardize the job titles by replacing "Mngr" and "Mng" with "Manager".

Answer:

Employee ID	Employee Name	Job Title
101	Alice Smith	Mngr
102	Bob Johnson	Manager
103	Charlie Lee	Mng
104	David Kim	Manager
105	Emma Watso	Mngr
106	Frank Martin	Manager
107	Grace Hall	Mng

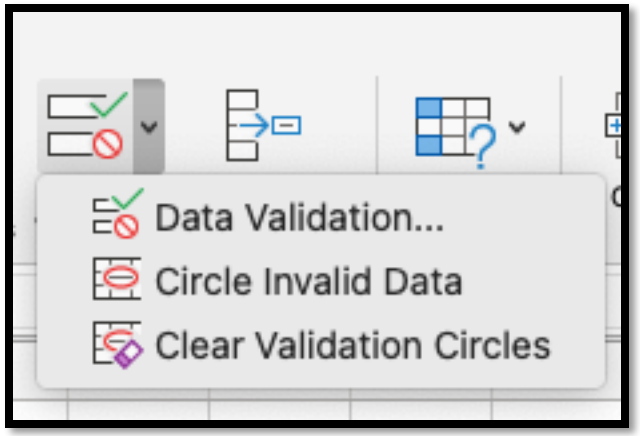
	A	B	C
1	Employee ID	Employee Name	Job Title
2	101	Alice Smith	Manager
3	102	Bob Johnson	Manager
4	103	Charlie Lee	Manager
5	104	David Kim	Manager
6	105	Emma Watson	Manager
7	106	Frank Martin	Manager
8	107	Grace Hall	Manager

d. Data Validation

- Allows to control the type of data that users can enter in a cell.
- It helps prevent errors, maintain data consistency, and ensure that the data entered meets specific criteria (e.g., numerical ranges, predefined lists, or specific formats).

Functions:

1. Restricting Data Entry to Specific Values
2. Creating Dropdown Lists
3. Setting Input Ranges or Rules
4. Custom Error Messages
5. Applying Validation to Entire Columns or Ranges



Scenario

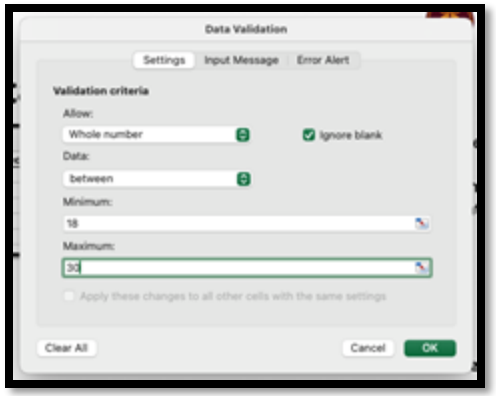
A university is managing a student registration form in Excel. They want to ensure that users enter valid data in certain fields to maintain consistency and prevent incorrect entries.

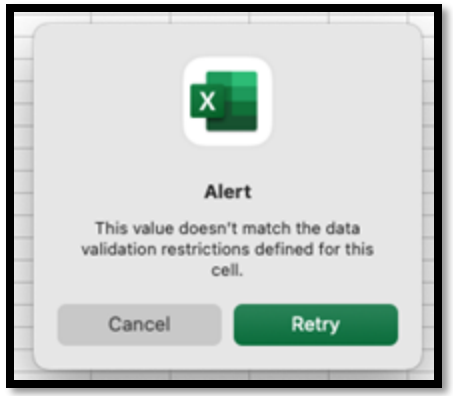
You will use **Data Validation** to implement the following rules:

1. Restrict the "Age" to be between **18 and 30**.
2. Create a dropdown list for "Course Selection" with the options: **"Computer Science," "Economics," "Business Administration"**.
3. Allow only email addresses in the "Email" field by setting custom criteria.

Source File: Data Validation-Student Reg

	A	B	C	D	E
1	Student ID	Student Name	Age	Course Selection	Email
2	101	Alice Smith			
3	102	Bob Johnson			
4	103	Charlie Lee			
5	104	Diana Kim			
6	105	Edward Martin			





If wrong input; error message will appear

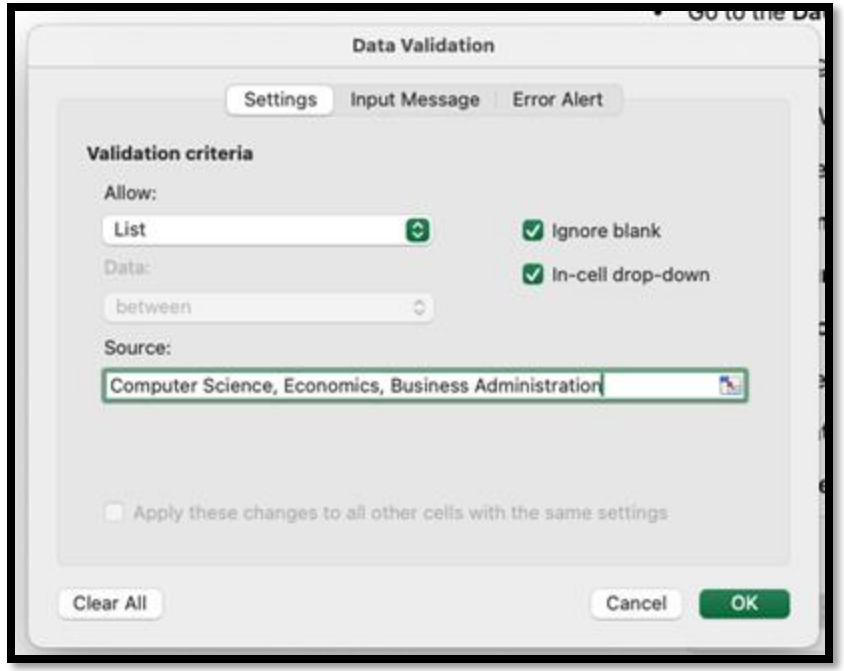
Apply the following **Data Validation** rules to the dataset:

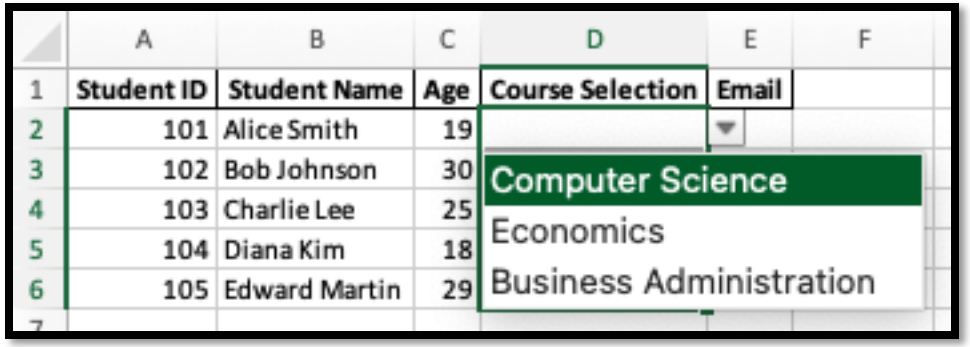
1.Age Restriction (18-30):

1. Select the cells in the **Age** column.
2. Go to the **Data** tab and click **Data Validation**.
3. In the Settings tab:
    1. **Allow:** Whole number
    2. **Data:** Between (18-30)

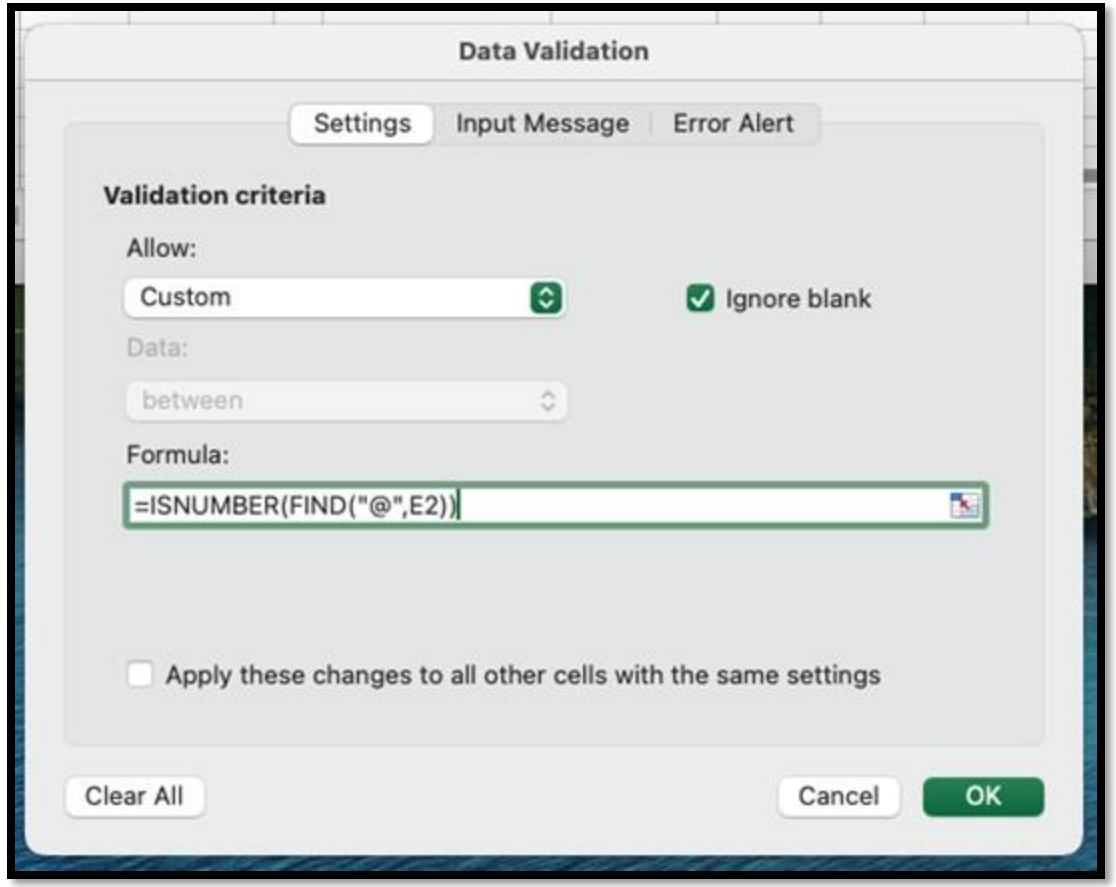
Repeated to Course Selection and Email

For Course Selection

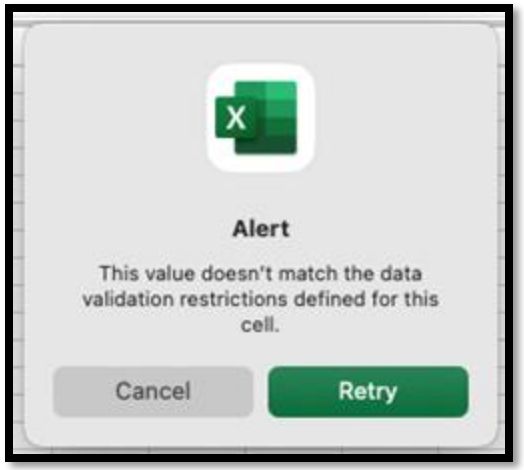




For Email



	A	B	C	D	E
1	Student ID	Student Name	Age	Course Selection	Email
2	101	Alice Smith	19	Computer Science	smith@utm.my
3	102	Bob Johnson	30	Economics	bob.my
4	103	Charlie Lee	25	Business Administration	
5	104	Diana Kim	18	Computer Science	
6	105	Edward Martin	29	Economics	





CLASS ACTIVITY

What are the formula for an email with criteria?

a. Must contain “@”,

b. Contain full stop “.”,

c. No empty space



What are the formula for an email with criteria?

Must contain “@”, Contain full stop “.”, No empty space

```
=IF(AND(ISNUMBER(FIND("@", E2)), ISNUMBER(FIND(".", E2)),
NOT(ISNUMBER(FIND(" ", E2)))), "Valid", "Invalid")
```

Explanation:

1.ISNUMBER(FIND("@", E2)): Checks if the @ symbol exists in the text of cell E2.

2.ISNUMBER(FIND(".", E2)): Checks if there is a period . in the email.

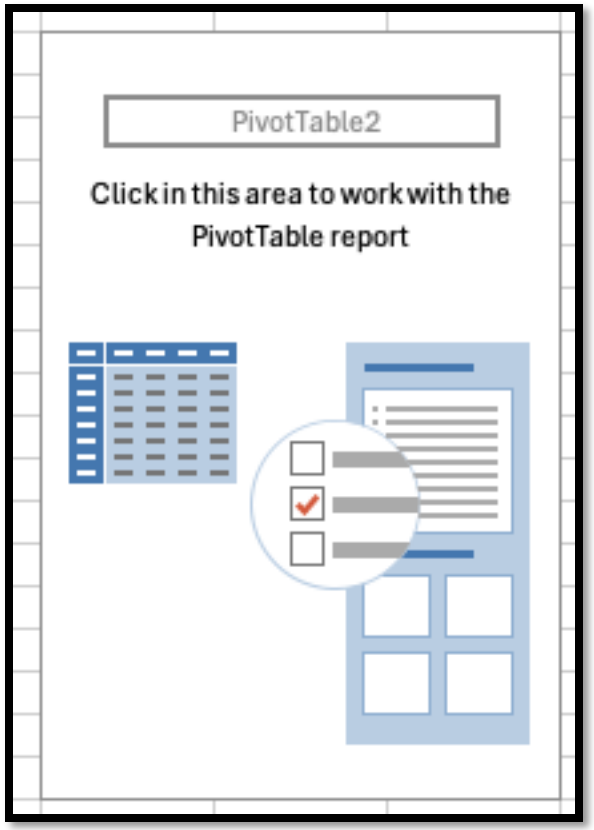
3.NOT(ISNUMBER(FIND(" ", E2))): Ensures there are no empty spaces in the email.

Functions for Data Validation

1. Restricting Data Entry to Specific Values
2. Creating Dropdown Lists
3. Setting Input Ranges or Rules
4. Custom Error Messages
5. Applying Validation to Entire Columns or Ranges

e. Pivot Table

- A **Pivot Table** used to summarize, analyze, and organize large sets of data.
- It helps users extract meaningful insights by automatically grouping, filtering, and rearranging data in a dynamic, interactive way.



Key Functions of a Pivot Table:

1. **Data Summarization**: Automatically summarizes data (e.g., sum, count, average).
2. **Data Filtering**: Filters specific categories or fields for focused analysis.
3. **Data Grouping**: Groups data by fields like dates, names, or categories.
4. **Dynamic Analysis**: Allows drag-and-drop flexibility to rearrange rows, columns, and values.
5. **Comparison**: Makes it easy to compare data across different variables.

Example

1. Declare Table

Insert -> Table

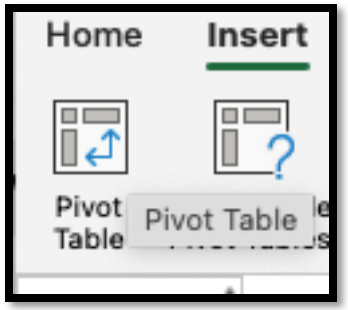


2. Highlight the range

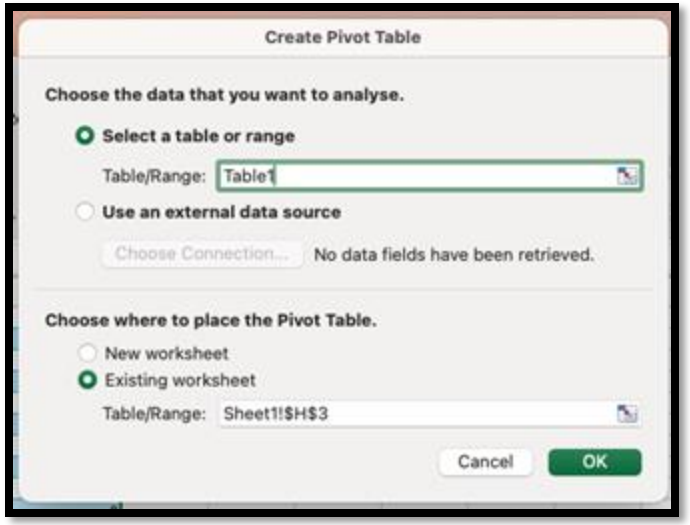


3. Create Pivot Table

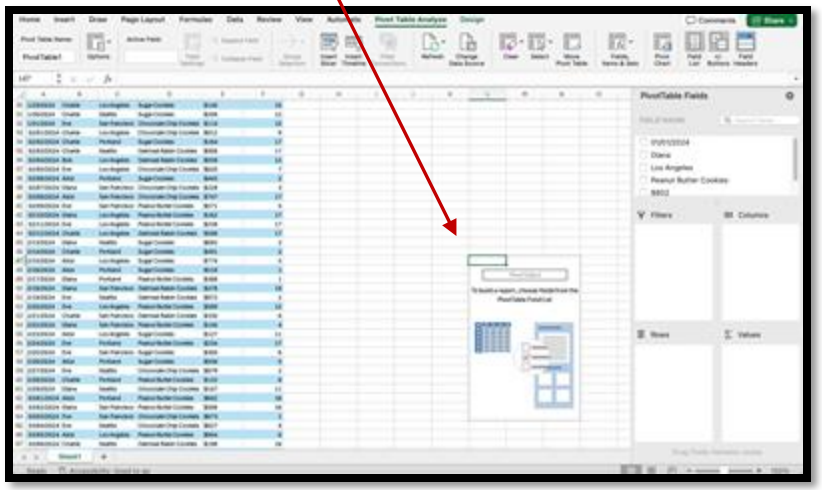
Insert Tab->Pivot Table



4. Pivot Table Location



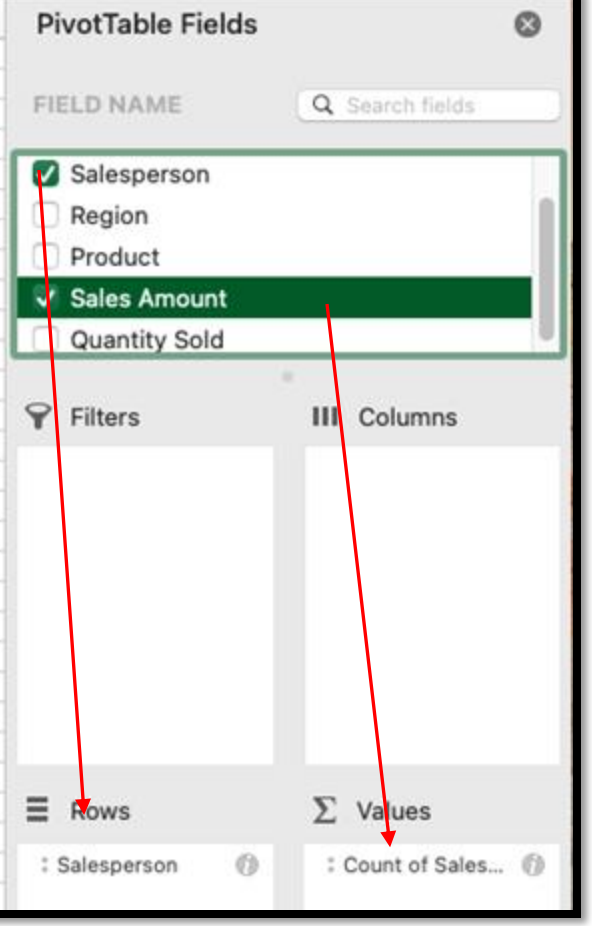
5. Pivot Table will appear



Salesperson (Rows) and Sales Amount (Sum)

	A	B	C	D	E	F
1	Date	Salesperson	Region	Product	Sales Amount	Quantity Sold
2	01/01/2024	Diana	Los Angeles	Peanut Butter Cookies	$802	7
3	01/02/2024	Alice	San Francisco	Sugar Cookies	$501	13
4	01/03/2024	Diana	San Francisco	Chocolate Chip Cookies	$829	15
5	01/04/2024	Charlie	Portland	Peanut Butter Cookies	$655	11
6	01/05/2024	Alice	San Francisco	Peanut Butter Cookies	$261	4
7	01/06/2024	Diana	Seattle	Sugar Cookies	$301	13
8	01/07/2024	Diana	Portland	Chocolate Chip Cookies	$369	7
9	01/08/2024	Charlie	San Francisco	Chocolate Chip Cookies	$962	19
10	01/09/2024	Alice	Portland	Peanut Butter Cookies	$915	2
11	01/10/2024	Charlie	Seattle	Oatmeal Raisin Cookies	$370	10
12	01/11/2024	Alice	Portland	Peanut Butter Cookies	$555	13
13	01/12/2024	Eve	Portland	Peanut Butter Cookies	$561	6
14	1/13/2024	Bob	Portland	Peanut Butter Cookies	$826	12
15	1/14/2024	Bob	Portland	Peanut Butter Cookies	$351	12
16	1/15/2024	Bob	Seattle	Sugar Cookies	$801	11
17	1/16/2024	Charlie	Los Angeles	Chocolate Chip Cookies	$395	7
18	1/17/2024	Eve	Portland	Sugar Cookies	$824	1
19	1/18/2024	Alice	Portland	Sugar Cookies	$819	1
20	1/19/2024	Diana	San Francisco	Sugar Cookies	$848	13
21	1/20/2024	Alice	Portland	Peanut Butter Cookies	$437	9
22	1/21/2024	Diana	San Francisco	Oatmeal Raisin Cookies	$978	3
23	1/22/2024	Alice	Los Angeles	Chocolate Chip Cookies	$152	7
24	1/23/2024	Eve	Seattle	Oatmeal Raisin Cookies	$891	6
25	1/24/2024	Diana	Seattle	Sugar Cookies	$316	8
26	1/25/2024	Charlie	Los Angeles	Sugar Cookies	$863	9
27	1/26/2024	Alice	Seattle	Oatmeal Raisin Cookies	$287	5
28	1/27/2024	Alice	Los Angeles	Oatmeal Raisin Cookies	$479	1
29	1/28/2024	Diana	Los Angeles	Oatmeal Raisin Cookies	$592	19

Row Labels	Count of Sales Amount
Alice	24
Bob	18
Charlie	20
Diana	21
Eve	17
Grand Total	100





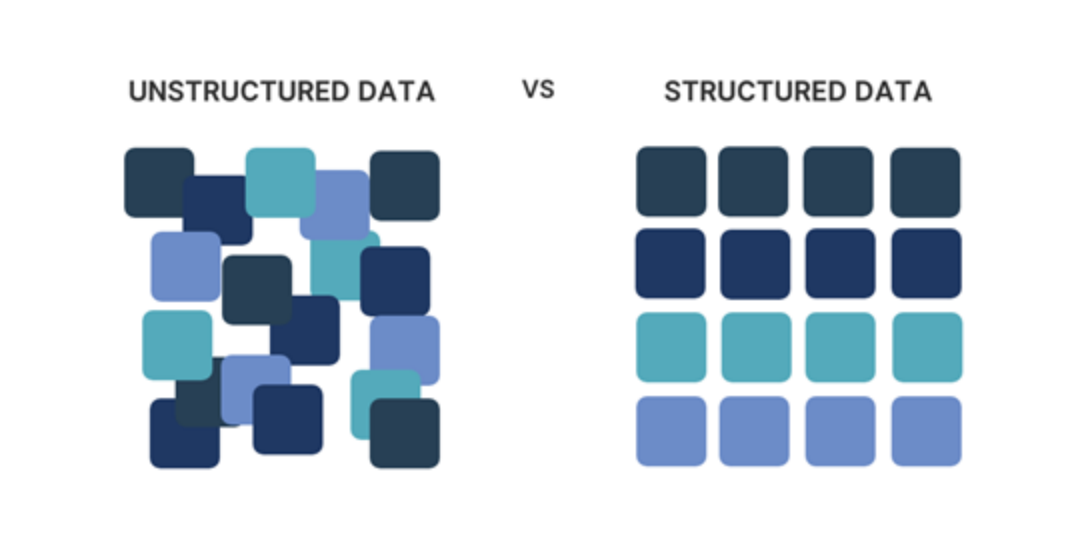
CLASS ACTIVITY

How about this table?

What are the selection?

Row Labels	Sum of Quantity Sold	Count of Sales Amount
Chocolate Chip Cookies	**158**	**20**
Alice	28	3
Bob	23	4
Charlie	35	3
Diana	36	4
Eve	36	6
Oatmeal Raisin Cookies	**264**	**26**
Alice	32	4
Bob	61	6
Charlie	105	8
Diana	53	5
Eve	13	3
Peanut Butter Cookies	**244**	**24**
Alice	51	7
Bob	40	3
Charlie	51	4
Diana	45	5
Eve	57	5
Sugar Cookies	**252**	**30**
Alice	69	10
Bob	58	5
Charlie	50	5
Diana	58	7
Eve	17	3
Grand Total	**918**	**100**

Data Cleaning



- Data cleaning improves the accuracy and reliability of datasets.
- Essential for quality data analysis and decision-making.
- Common issues include missing data, duplicates, and inconsistent formatting.
- There are two type of data cleaning, basic and advanced.

Data Cleaning Techniques

1. **Handling Missing Data** – Remove or impute missing values.
2. **Removing Duplicates** – Avoid redundant data.
3. **Correcting Inconsistencies** – Standardize formats, fix typos.
4. **Handling Outliers** – Identify and transform extreme values.
5. **Normalization & Standardization** – Rescale numerical data.
6. **Validating Data Accuracy** – Cross-check with reliable sources.
7. **Removing Irrelevant Data** – Filter out unnecessary information.

a. Basic Functions for Data Cleaning

- **IF()** – Compare value or identify and handle missing values
- **COUNTBLANK()** – Count missing values in a range
- **TRIM()** – Remove unwanted spaces
- **CLEAN()** – Remove non-printable characters
- **ISBLANK()** – Check for empty cells

IF - Handle missing values

Dataset: IF

Student	Score
Alice	85
Bob	
Charlie	78
David	90
Emma	
Frank	65

Answer:

Student	Score	Output
Alice	85	85
Bob		Missing
Charlie	78	78
David	90	90
Emma		Missing
Frank	65	65

Scenario:

You are given a dataset of student scores, but some entries are missing. Your task is to use the IF() function to identify missing values and label them as **"Missing"** while keeping valid scores unchanged.

Instructions:

1.Open **Excel** and enter the above data into columns **A** and **B**.

2.In **Column C**, use the IF() function to check if the score is missing:

- If the score is blank, return "Missing".
- Otherwise, return the score itself.

Formula to Use in C2:

=IF(B2="", "Missing", B2)

3. Drag the formula down to apply it to all rows.
4. Observe how Excel replaces missing values with the text "Missing".

IF- Compare value or identify

Dataset: Compare Budget

	A	B	C	D
1	Item	Budget ($)	Actual Expense ($)	Status (Formula)
2	Rent	1000	1200	
3	Utilities	200	180	
4	Marketing	500	600	
5	Salaries	3000	2900	
6	Office Suppli	150	150	
7	Travel	400	500	

Answer:

	A	B	C	D
1	Item	Budget ($)	Actual Expense ($)	Status (Formula)
2	Rent	1000	1200	Over Budget
3	Utilities	200	180	Within Budget
4	Marketing	500	600	Over Budget
5	Salaries	3000	2900	Within Budget
6	Office Supp	150	150	Within Budget
7	Travel	400	500	Over Budget

Scenario:

A company tracks expenses against a set budget. You need to use the IF() function to determine whether each expense is **"Over Budget"** or **"Within Budget"**.

Instructions:

1. Open **Excel** and enter the above data into **Columns A, B, and C**.
2. In **Column D (Status)**, use the IF() function to check if the actual expense is greater than the budget:
   - If the actual expense is greater than the budget, return "Over Budget".
   - Otherwise, return "Within Budget".

   Formula to Use in D2:

   =IF(C2>B2, "Over Budget", "Within Budget")

1. Drag the formula down to apply it to all rows.
2. Observe how Excel labels each item as either **"Over Budget"** or **"Within Budget"**.

COUNTBLANK() - Count missing values in a range

Dataset: Employee Attendance Record

	A	B	C	D	E	F
1	Employee Name	Day 1	Day 2	Day 3	Day 4	Day 5
2	Alice	P	P		P	P
3	Bob	P		P	P	P
4	Charlie	P	P	P		
5	Diana		P	P	P	P
6	Edward	P	P	P	P	

Answer:

	A	B	C	D	E	F	G	H
1	Employee Name	Day 1	Day 2	Day 3	Day 4	Day 5		
2	Alice	P	P		P	P		
3	Bob	P		P	P	P	1	Day 1
4	Charlie	P	P	P			2	Day 5
5	Diana		P	P	P	P		
6	Edward	P	P	P	P			

Scenario:

A company tracks employee attendance in a monthly sheet. Some employees forgot to mark their attendance, leaving blank spaces in the sheet. Your task is to count the number of blank entries to identify missing records.

Instructions:

1. Find the column with employee attendance records. Blank cells indicate missing entries.

Select the Range

2. Click on an empty cell where you want to display the count of missing values.
3. For example, if attendance data is in cells **B2:B20**, select a blank cell (e.g., B2).
4. Apply the COUNTBLANK() Function

   =COUNTBLANK(B2:B20)

TRIM() - Remove unwanted spaces

Dataset: Customer List with Extra Spaces

	A	B	C	D	E
1	Customer ID	Customer Name (Original)	Phone Number (Original)	Trimmed Name (Formula)	Trimmed Phone (Formula)
2	101	John Doe	123-456-7890		
3	102	Jane Smith	987-654-3210		
4	103	Michael Lee	555-666-7777		
5	104	Emma Watson	333-222-1111		
6	105	Alice Brown	777-888-9999		
7					

Answer:

	A	B	C	D	E
1	Customer ID	Customer Name (Original)	Phone Number (Original)	Trimmed Name (Formula)	Trimmed Phone (Formula)
2	101	John Doe	123-456-7890	John Doe	123-456-7890
3	102	Jane Smith	987-654-3210	Jane Smith	987-654-3210
4	103	Michael Lee	555-666-7777	Michael Lee	555-666-7777
5	104	Emma Watson	333-222-1111	Emma Watson	333-222-1111
6	105	Alice Brown	777-888-9999	Alice Brown	777-888-9999
7					

Scenario:

A company maintains a customer database, but some entries have extra spaces before or after the text. This affects sorting, searching, and data consistency. Your task is to clean up the text data using the TRIM() function in Excel.

Instructions:

1. Select a Blank Column for Cleaned Data
2. In **Column D (Cleaned Name)**, you will apply the TRIM() function to remove extra spaces from **Column B (Customer Name)**.
3. Click on **Cell D2** and type the following formula:
   =TRIM(B2)
4. Click on the **bottom-right corner** of **Cell D2** and drag it down to apply the formula to all rows.
5. Compare **Column B (Original Data)** with **Column D (Cleaned Data)** to see how the extra spaces have been removed.

Can you please trimmed the Column E?

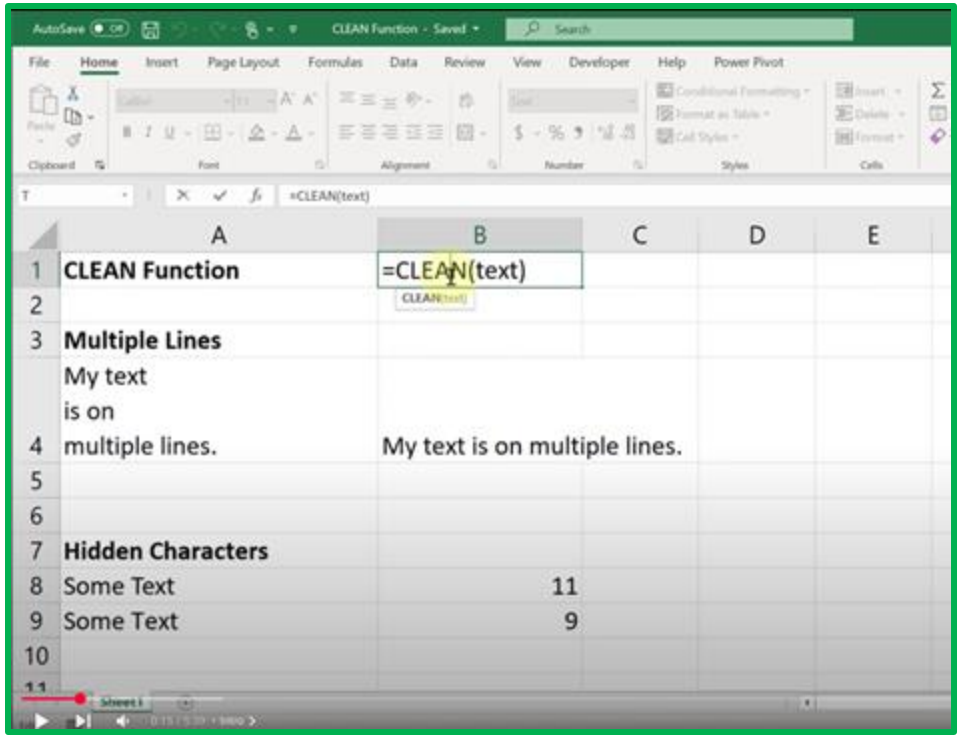
www.fai.utm.my

CLASS ACTIVITY - CLEAN()

YouTube

https://www.youtube.com/watch?v=3K7jw_c6SeY

Duration = 5:39 minutes



ISBLANK()- Check for empty cells

Dataset: Student Grades

	A	B	C	D	E
1	Student ID	Student Name	Math Score	English Score	Status (Formula)
2	101	Alice	85		
3	102	Bob		76	
4	103	Charlie	78	89	
5	104	Diana	90		
6	105	Edward		92	
7	106	Fiona	88		
8	107	George		80	

Answer:

	A	B	C	D	E	F
1	Student ID	Student Name	Math Score	English Score	Status (Formula)	
2	101	Alice	85		FALSE	TRUE
3	102	Bob		76	TRUE	FALSE
4	103	Charlie	78	89	FALSE	FALSE
5	104	Diana	90		FALSE	TRUE
6	105	Edward		92	TRUE	FALSE
7	106	Fiona	88		FALSE	TRUE
8	107	George		80	TRUE	FALSE

Scenario:

A school is tracking student grades for different subjects. However, some students have not completed their exams, and their scores are missing from the dataset. Your task is to use the ISBLANK() function to identify missing scores and flag students with incomplete records.

Instructions:

1. In **Column E (Status)**, use the ISBLANK() function to flag missing Math scores:
2. Click on **Cell E2** and type the formula

   =ISBLANK(C2)

3. Drag the formula from **E2** down to **E8** to apply it to all students to apply to other rows.

b. Advanced Functions for Data Cleaning

- **VLOOKUP() / XLOOKUP()** – Fill missing values from another dataset
- **TEXTJOIN()** – Merge data while handling missing values
- **IFERROR()** – Replace errors with meaningful values
- **ARRAYFORMULA()** – Automate data handling across rows
- **Power Query (Excel)** – Automate data cleaning workflows

VLOOKUP()- Fill missing values from another dataset

Function of VLOOKUP in Excel

VLOOKUP() (**Vertical Lookup**) is a function in Excel used to **search for a value in one column and return a corresponding value from another column in the same row**. It is commonly used to retrieve information from large datasets, such as product prices, employee details, or student grades.

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

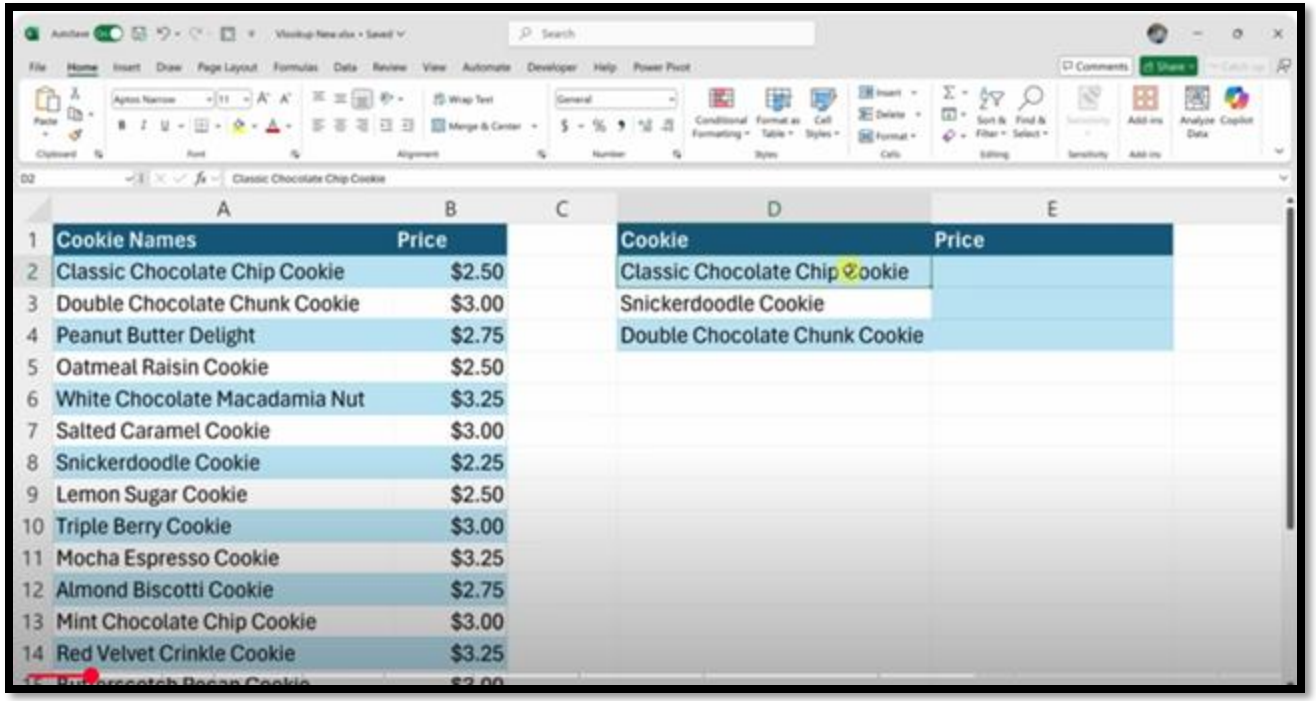
Arguments:

1. **lookup_value** – The value you want to search for (e.g., Product ID, Employee ID).
2. **table_array** – The range of data containing the lookup value and the value to return.
3. **col_index_num** – The column number (starting from 1) from which to return the result.
4. **[range_lookup]** – (Optional)
   - TRUE (Approximate match) – If an exact match is not found, it returns the closest lower value.
   - FALSE (Exact match) – Only returns a result if an exact match is found.

CLASS ACTIVITY - VLOOKUP()

YouTube

Duration = 5:42 minutes



	A	B	C	D	E
1	Cookie Names	Price		Cookie	Price
2	Classic Chocolate Chip Cookie	$2.50		Classic Chocolate Chip Cookie	
3	Double Chocolate Chunk Cookie	$3.00		Snickerdoodle Cookie	
4	Peanut Butter Delight	$2.75		Double Chocolate Chunk Cookie	
5	Oatmeal Raisin Cookie	$2.50			
6	White Chocolate Macadamia Nut	$3.25			
7	Salted Caramel Cookie	$3.00			
8	Snickerdoodle Cookie	$2.25			
9	Lemon Sugar Cookie	$2.50			
10	Triple Berry Cookie	$3.00			
11	Mocha Espresso Cookie	$3.25			
12	Almond Biscotti Cookie	$2.75			
13	Mint Chocolate Chip Cookie	$3.00			
14	Red Velvet Crinkle Cookie	$3.25			

https://www.youtube.com/watch?v=xlynD1gFOLo

CLASS ACTIVITY - VLOOKUP()

Scenario:

You are tasked with finding specific details about employees based on their Employee ID using VLOOKUP.

Fill in the missing cells under "Output" by looking up the Employee ID in the main data table and returning the corresponding Name, Department, and Salary.

	A	B	C	D
1	Employee ID	Name	Department	Salary
2	Column1	Column2	Column3	Column4
3	E101	Employee_1	Sales	40500
4	E102	Employee_2	HR	41000
5	E103	Employee_3	IT	41500
6	E104	Employee_4	Finance	42000
7	E105	Employee_5	Marketing	42500
8	E106	Employee_6	Sales	43000
9	E107	Employee_7	HR	43500
10	E108	Employee_8	IT	44000
11	E109	Employee_9	Finance	44500
12	E110	Employee_10	Marketing	45000
13	E111	Employee_11	Sales	45500
14	E112	Employee_12	HR	46000
15	E113	Employee_13	IT	46500
16	E114	Employee_14	Finance	47000
17	E115	Employee_15	Marketing	47500
18	E116	Employee_16	Sales	48000
19	E117	Employee_17	HR	48500
20	E118	Employee_18	IT	49000
21	E119	Employee_19	Finance	49500
22	E120	Employee_20	Marketing	50000
23	E121	Employee_21	Sales	50500
24	E122	Employee_22	HR	51000
25	E123	Employee_23	IT	51500
26	E124	Employee_24	Finance	52000
27	E125	Employee_25	Marketing	52500
28	E126	Employee_26	Sales	53000
29	E127	Employee_27	HR	53500
30	E128	Employee_28	IT	54000
31	E129	Employee_29	Finance	54500
32	E130	Employee_30	Marketing	55000
33	E131	Employee_31	Sales	55500
34	E132	Employee_32	HR	56000
35	E133	Employee_33	IT	56500
36	E134	Employee_34	Finance	57000
37	E135	Employee_35	Marketing	57500
38	E136	Employee_36	Sales	58000
39	E137	Employee_37	HR	58500
40	E138	Employee_38	IT	59000
41	E139	Employee_39	Finance	59500
42	E140	Employee_40	Marketing	60000
43	E141	Employee_41	Sales	60500
44	E142	Employee_42	HR	61000
45	E143	Employee_43	IT	61500
46	E144	Employee_44	Finance	62000
47	E145	Employee_45	Marketing	62500
48	E146	Employee_46	Sales	63000
49	E147	Employee_47	HR	63500
50	E148	Employee_48	IT	64000
51	E149	Employee_49	Finance	64500
52	E150	Employee_50	Marketing	65000

Task:

01

Department of the employee with Employee 30.

02

Salary of the employee with Employee ID E144.

03

Name of the employee with Employee ID E145

Output:

Info	Question	Answer
Department of the employee with **Employee 30**	Employee_30	Marketing
Salary of the employee with **Employee ID E144**	E130	55000
Name of the employee with **Employee ID E145**	E145	Employee_45

TEXTJOIN() - Merge data while handling missing values

Dataset:

	A	B	C
1	First Name	Last Name	Student ID
2	John	Smith	1
3	Maria	Lopez	2
4	Ahmed	Khan	3
5	Sarah	Brown	4
6	Priya	Patel	5

Scenario:

You work at a university's administrative office, and you need to create a system that combines student details into a single string for easier data processing. The data includes **First Name**, **Last Name**, and **Student ID**. You want to combine these details with specific delimiters (like spaces, commas, or hyphens) to generate a comprehensive record for each student.

Answer:

	A	B	C	D
1	First Name	Last Name	Student ID	
2	John	Smith	1	John-Smith-1
3	Maria	Lopez	2	Maria-Lopez-2
4	Ahmed	Khan	3	Ahmed-Khan-3
5	Sarah	Brown	4	Sarah-Brown-4
6	Priya	Patel	5	Priya-Patel-5

Instructions:

Use the **TEXTJOIN** function to create a string that combines the student's first name, last name, and student ID, separated by hyphens (-).

1. Enter the following formula in a new column (D2), and drag it down for all students:

   =TEXTJOIN("-", TRUE, A2, B2, C2)

CLASS ACTIVITY - TEXTJOIN()

You are working in a customer service team and need to clean up and consolidate incomplete customer data into a single cell for easy reference.

The database contains multiple columns, such as **First Name**, **Last Name**, **Phone Number**, and **Email**.

Some data entries are incomplete (i.e., some cells are blank), and you want to generate a string that includes only the non-empty values.

	A	B	C	D
1	**First Name**	**Last Name**	**Phone Number**	**Email Address**
2	Emily	Johnson	123-456-7890	emilyj@example.com
3	Michael	Lee		michaell@example.com
4	Ahmed	Khan	987-654-3210	
5	Sarah		654-987-1234	sarahb@example.com
6	Priya	Patel		

Answer:

	A	B	C	D	E
1	**First Name**	**Last Name**	**Phone Number**	**Email Address**	
2	Emily	Johnson	123-456-7890	emilyj@example.com	Emily, Johnson, 123-456-7890, emilyj@example.com
3	Michael	Lee		michaell@example.com	Michael, Lee, michaell@example.com
4	Ahmed	Khan	987-654-3210		Ahmed, Khan, 987-654-3210
5	Sarah		654-987-1234	sarahb@example.com	Sarah, 654-987-1234, sarahb@example.com
6	Priya	Patel			Priya, Patel

IFERROR()- Replace errors with meaningful values

Scenario:

- You are working as a financial analyst and need to calculate the **profit margin** for a list of products.

- However, some products have missing sales or cost data, causing errors (like #DIV/0!) when performing calculations.

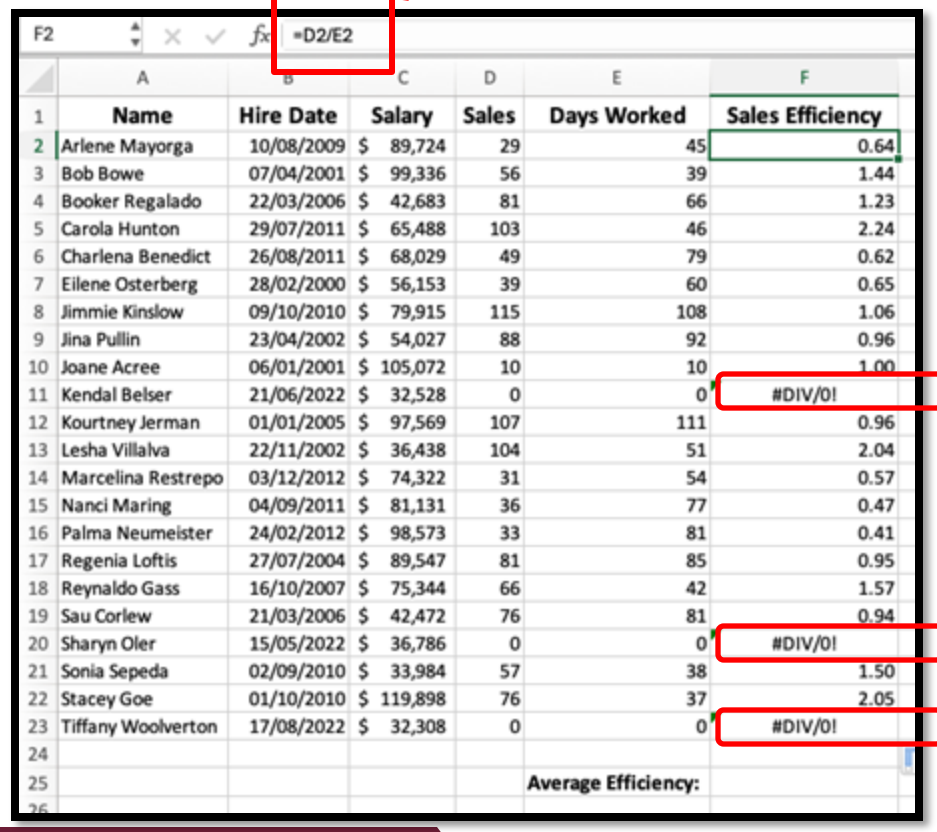
- To clean up the spreadsheet and make it more user-friendly, you will use the **IFERROR** function to replace error messages with more meaningful outputs (e.g., "Incomplete Data").

Data Example in Excel:

Scenario: Identify
(i) Sales Efficiency (Sales/Days Worked)
(ii) Average Efficiency

If in F2 the value =D2/E2 will produce the value of Sales Efficiency

Output:



	A	B	C	D	E	F
1	Name	Hire Date	Salary	Sales	Days Worked	Sales Efficiency
2	Arlene Mayorga	10/08/2009	$ 89,724	29	45	0.64
3	Bob Bowe	07/04/2001	$ 99,336	56	39	1.44
4	Booker Regalado	22/03/2006	$ 42,683	81	66	1.23
5	Carola Hunton	29/07/2011	$ 65,488	103	46	2.24
6	Charlena Benedict	26/08/2011	$ 68,029	49	79	0.62
7	Eilene Osterberg	28/02/2000	$ 56,153	39	60	0.65
8	Jimmie Kinslow	09/10/2010	$ 79,915	115	108	1.06
9	Jina Pullin	23/04/2002	$ 54,027	88	92	0.96
10	Joane Acree	06/01/2001	$ 105,072	10	10	1.00
11	Kendal Belser	21/06/2022	$ 32,528	0	0	#DIV/0!
12	Kourtney Jerman	01/01/2005	$ 97,569	107	111	0.96
13	Lesha Villalva	22/11/2002	$ 36,438	104	51	2.04
14	Marcelina Restrepo	03/12/2012	$ 74,322	31	54	0.57
15	Nanci Maring	04/09/2011	$ 81,131	36	77	0.47
16	Palma Neumeister	24/02/2012	$ 98,573	33	81	0.41
17	Regenia Loftis	27/07/2004	$ 89,547	81	85	0.95
18	Reynaldo Gass	16/10/2007	$ 75,344	66	42	1.57
19	Sau Corlew	21/03/2006	$ 42,472	76	81	0.94
20	Sharyn Oler	15/05/2022	$ 36,786	0	0	#DIV/0!
21	Sonia Sepeda	02/09/2010	$ 33,984	57	38	1.50
22	Stacey Goe	01/10/2010	$ 119,898	76	37	2.05
23	Tiffany Woolverton	17/08/2022	$ 32,308	0	0	#DIV/0!
24						
25					Average Efficiency:	

Without IFERROR, your formulas in the ”Sales Efficiency Calculation" column might return errors like #DIV/0! when dividing by zero or #VALUE! if the sales or data worked is missing.

How to solve?

	A	B	C	D	E	F
1	Name	Hire Date	Salary	Sales	Days Worked	Sales Efficiency
2	Arlene Mayorga	10/08/2009	$ 89,724	29	45	0.64
3	Bob Bowe	07/04/2001	$ 99,336	56	39	1.44
4	Booker Regalado	22/03/2006	$ 42,683	81	66	1.23
5	Carola Hunton	29/07/2011	$ 65,488	103	46	2.24
6	Charlena Benedict	26/08/2011	$ 68,029	49	79	0.62
7	Eilene Osterberg	28/02/2000	$ 56,153	39	60	0.65
8	Jimmie Kinslow	09/10/2010	$ 79,915	115	108	1.06
9	Jina Pullin	23/04/2002	$ 54,027	88	92	0.96
10	Joane Acree	06/01/2001	$ 105,072	10	10	1.00
11	Kendal Belser	21/06/2022	$ 32,528	0	0	0.00
12	Kourtney Jerman	01/01/2005	$ 97,569	107	111	0.96
13	Lesha Villalva	22/11/2002	$ 36,438	104	51	2.04
14	Marcelina Restrepo	03/12/2012	$ 74,322	31	54	0.57
15	Nanci Maring	04/09/2011	$ 81,131	36	77	0.47
16	Palma Neumeister	24/02/2012	$ 98,573	33	81	0.41
17	Regenia Loftis	27/07/2004	$ 89,547	81	85	0.95
18	Reynaldo Gass	16/10/2007	$ 75,344	66	42	1.57
19	Sau Corlew	21/03/2006	$ 42,472	76	81	0.94
20	Sharyn Oler	15/05/2022	$ 36,786	0	0	0.00
21	Sonia Sepeda	02/09/2010	$ 33,984	57	38	1.50
22	Stacey Goe	01/10/2010	$ 119,898	76	37	2.05
23	Tiffany Woolverton	17/08/2022	$ 32,308	0	0	0.00
24						
25					Average Efficiency:	0.97

Instructions:

1. In cell F2 use formula:
   =IFERROR(D2/E2,0)
2. Drag it down for the other rows.

Question: Average Efficiency ?

=

ARRAYFORMULA() - Automate data handling across rows

Scenario:

- You are managing inventory for an online electronics store.
- You want to calculate the **total value of stock** for different categories of items (e.g., Laptops, Smartphones, Accessories) by multiplying the stock quantity by the price for each item.
- Instead of manually calculating and summing the values for each row, you will use an **array formula** to streamline the process.



Data

	A	B	C	D
1	**Item**	**Category**	**Stock Quantity**	**Price per Item ($)**
2	Laptop 1	Laptops	50	800
3	Smartphone 1	Smartphones	100	500
4	Accessory 1	Accessories	200	20
5	Laptop 2	Laptops	30	850
6	Smartphone 2	Smartphones	150	450
7	Accessory 2	Accessories	300	15

Non-Array formula:

1. E2 =C2*D2
2. D8 =sum(E2:E7)

Array Formula

1. E1 =SUM(C2:C7 * D2:D7)
2. Press **Ctrl + Shift + Enter**
3. Curly braces {} will automatically appear around your formula:

   {=SUM(C2:C7 * D2:D7)}



Sales Performance Analysis

You are part of a sales analysis team tasked with summarizing sales data for a company that operates in five regions. Your goal is to analyze the data using array formulas to calculate total sales, average sales, and high-performing sales regions.

Region	January Sales ($)	February Sales ($)	March Sales ($)	April Sales ($)	May Sales ($)	June Sales ($)
North	12,000	15,000	18,000	20,000	25,000	28,000
South	10,000	12,000	14,000	16,000	20,000	24,000
East	8,000	9,000	11,000	12,000	15,000	18,000
West	14,000	16,000	17,000	21,000	26,000	30,000
Central	11,000	13,000	16,000	18,000	22,000	25,000

Questions

1. Calculate the total sales for all months and regions.
2. Calculate Average Sales Per Month for Each Region.

CLASS ACTIVITY (Basic & Advanced)

Dataset: Employee Information

	A	B	C	D	E	F
1	**Employee ID**	**Name**	**Age**	**Email**	**Salary**	**Remarks**
2	101	John Doe	25	johndoe@email.com	50,000	(space)
3	102	Jane Doe		janed@email.com	55,500	#ERROR
4	103	Sam Smith	30	samsmith@email.com	60,000	Good
5	104	Lily Lee	27		45,000	Excellent
6	105	Mark Taylor	22	marktaylor@email	48,750	Outstanding
7	106		29	ann@gmail.com	52,000	(extra spaces)
8	107	Alex Chen	31	alex.chen@email.com		Great
9	108	Sarah Kim	24	sarahkim@email.com	53,250	#N/A
10	109	Mike Brown		mikebrown@email.com	49,800	Good

Instruction: Apply the following Excel functions to clean the data.

1. **IF(ISBLANK(B2:B10),** "Missing", B2:B10) → Replace missing names
2. **COUNTBLANK(C2:C10)** → Count missing ages
3. **TRIM(F2:F10)** → Remove extra spaces in remarks
4. **CLEAN(F2:F10)** → Remove non-printable characters
5. **IFERROR(F2:F10, "Check for Errors")** → Replace error values
6. **PROPER(B2:B10)** → Capitalize names correctly
7. **IF(ISBLANK(D2:D10),** "Email Missing", D2:D10) → Identify missing emails

Power Query (Excel)

- Power Query is a data connection technology that allows users to import, clean, and transform data efficiently.
- Power Query is a widely used ETL (Extract, Transform, Load) tool.
- It helps in automating data preparation tasks and enables seamless integration from multiple sources.
- Key benefits:
  a. Easy data transformation without coding
  b. Automates repetitive tasks
  c. Works with multiple data sources (Excel, databases, online services, etc.)

Key Features and Functions

a. **Import Data:** Extract data from various sources (Excel, SQL, Web, etc.).

b. **Transform Data:** Remove duplicates, change formats, split/merge columns.

c. **Merge & Append Queries:** Combine data from multiple tables efficiently.

d. **Custom Columns & Formulas:** Use M language for advanced transformations.

e. **Load Data:** Save processed data into Excel or Power BI for further analysis.

Steps to Use Power Query

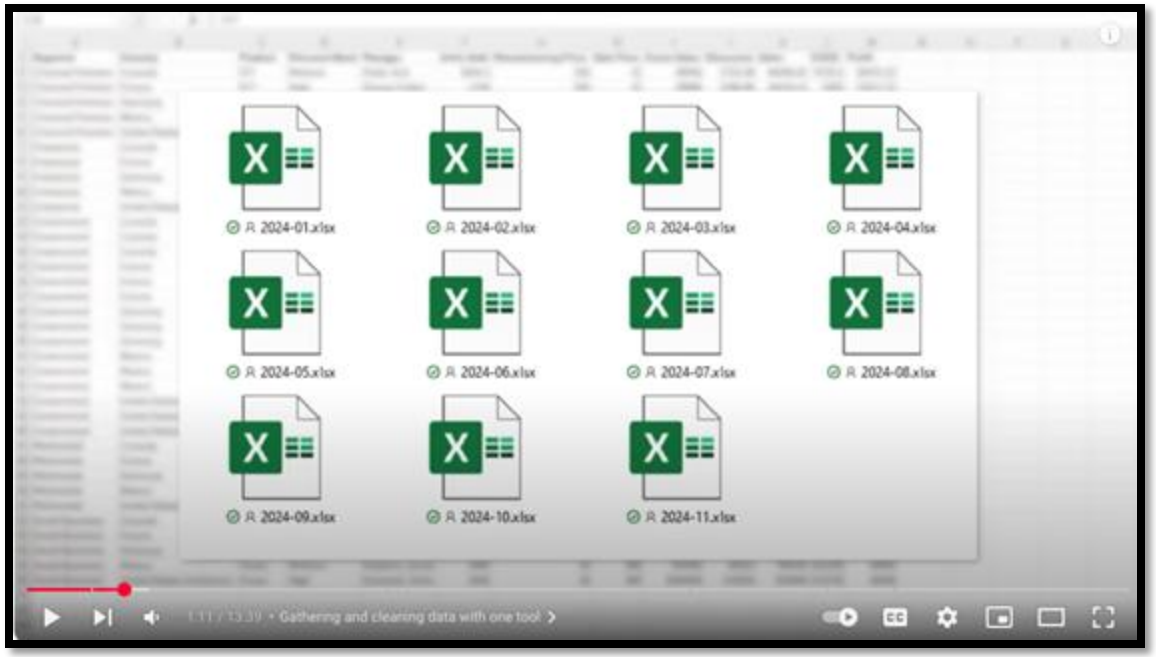
1. **Open Power Query:** Go to Data tab > Get & Transform > Get Data.
2. **Connect to Data Source:** Choose source (Excel, Web, Database, etc.).
3. **Transform Data:** Apply filters, clean, and structure data as needed.
4. **Load Data:** Click Close & Load to import cleaned data into Excel.

Tip: Use the **Power Query Editor** for advanced data shaping!

Power Query (Excel)

YouTube

Duration = 13:39 minutes



Data Analysis for BEGINNERS in Excel in 2025 | No Python Required

https://www.youtube.com/watch?v=wSFWJ0AF7SM

KNOWLEDGE CHECK



THANK YOU



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